

USTHB University
Faculty of Computer Science
AI&DS Department

جامعة هواري بومدين

Internet of Things (IoT)

01- Introduction to IoT



5th Engineers | AMEUR IMANE | 2026/2027

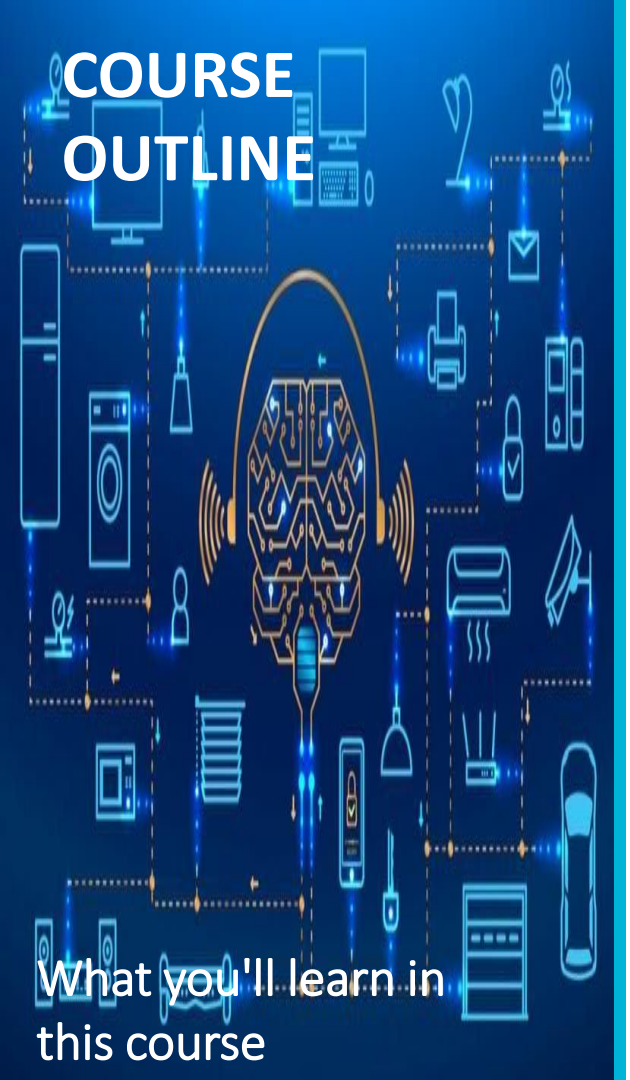
Internet of Thing (IoT):
Engineers (5th Year) : Software Engineering

5- Semestre 9 :

Unité d'Enseignement	VHS	V.H hebdomadaire				Coeff.	Crédits	Mode d'enseignement		Mode d'évaluation	
	14-16 sem	C	TD	TP	Autres			A distance	En présentiel	Continu	Examen
UE fondamentales											
UEF1(O/P)											
Méthodes formelles pour GL	42h	1h30	1h30/15J	1h30/15J		3	4		X	40%	60%
Développement de logiciels embarqués	42h	1h30		1h30		3	4		X	40%	60%
UEF2(O/P)											
Conception de jeux vidéo : Théorie et Pratique	42h	1h30		1h30		3	4		X	40%	60%
Internet of Things (IoT) : Concepts et développement	63h	1h30	1h30	1h30		3	5		X	40%	60%
UE méthodologie											
UEM1 (O/P)											
DevOPs & Cloud Computing	84h	3h00		3h00		3	5		X	40%	60%
Sécurité Logicielle	42h	1h30		1h30		2	3		X	40%	60%
Développement mobile	42h	1h30		1h30		2	3			40%	60%
UE transversales											
UET1(O/P)											
Aspects juridiques	21h	1h30				1	2		X		100%
Total Semestre 9	**378h**	**13h30**	**2h15**	**11h15**		**20**	**30**				

- Volume hebdomadaire = 27H00 / semaine.
- 8 Matières par semestre.

COURSE OUTLINE



What you'll learn in this course

01 **Definitions**
IoT and connected objects

02 **Characteristics**
What defines an IoT system

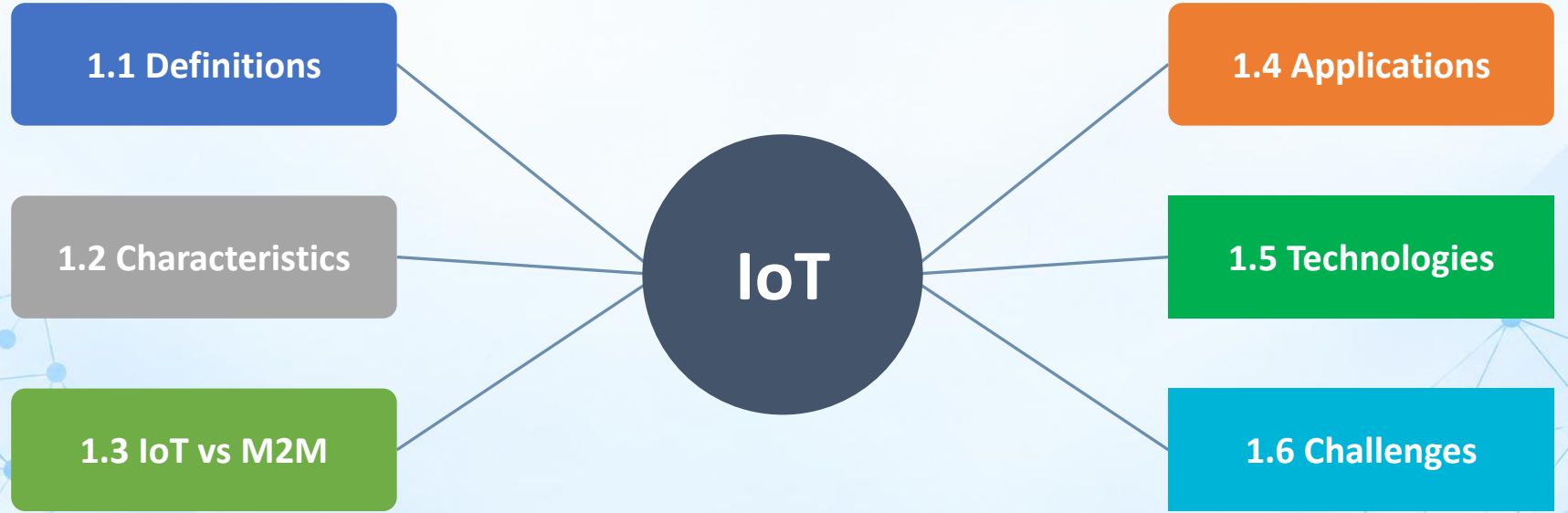
03 **IoT vs M2M**
From mainframes to the cloud

04 **Application domains**
Home, city, health, agriculture, industry

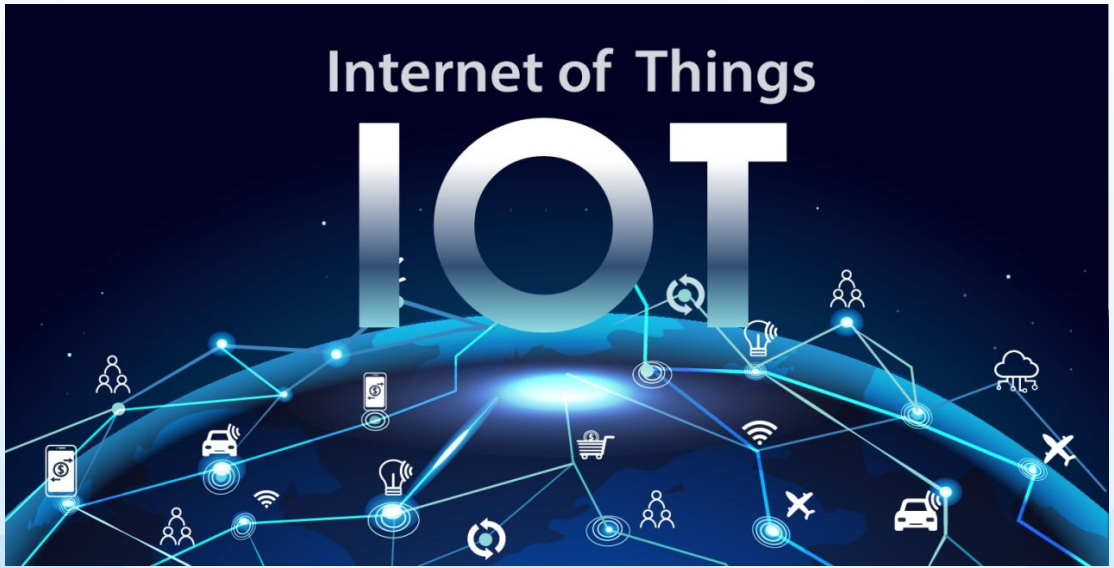
05 **Enabling technologies**
WSN, Cloud, Big Data, protocols, embedded systems

06 **Challenges**
Security, energy, interoperability...

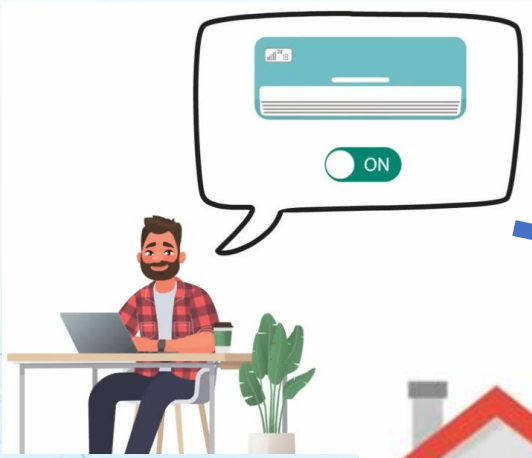
Overview: Chapter Mind Map



1. Definitions



Context and motivation



IoT







Context and motivation









Smart Environment can be defined as making users interact easily with their immediate surroundings due to the invention of intelligent technologies, attached with software-based services.





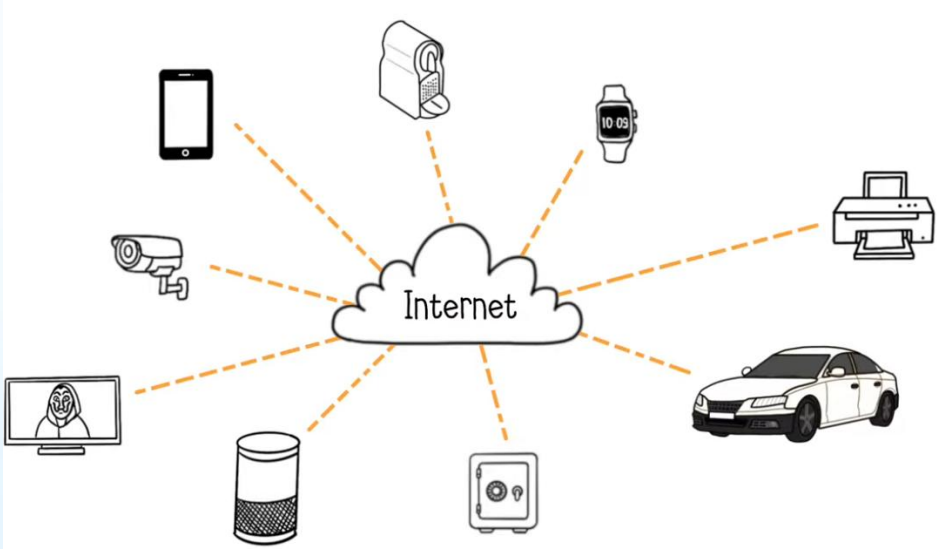


Context and motivation

IoT is a concept, not a specific technology or device.

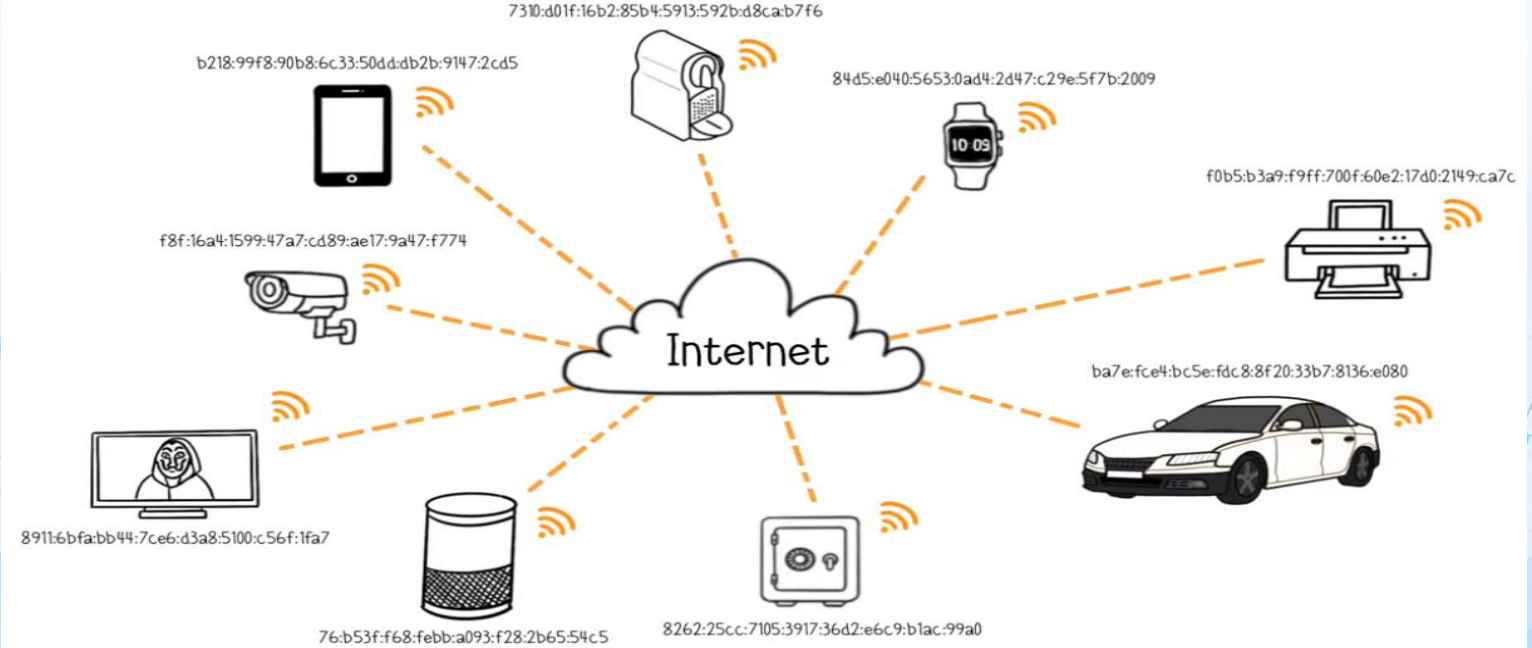


Internet



Context and motivation

Each connected object has a standardized digital identity, such as an IP address, and communicates via wireless technologies like RFID, Bluetooth, or Wi-Fi.
These objects are also equipped with sensors that collect the data.



What is the Internet of Things?

A network of physical devices connected to the internet

It allows objects to be sensed and controlled remotely across existing network infrastructure

Also called The Internet of Objects and IOT



What is the Internet of Things?

It allows communication between human and things

Implemented by embedding sensors in different objects that can transmit and receive data

Makes everyday life smarter, healthier, and more efficient



Three complementary visions of IoT

“Things” vision
Identify and track objects: RFID, barcodes, smart tags

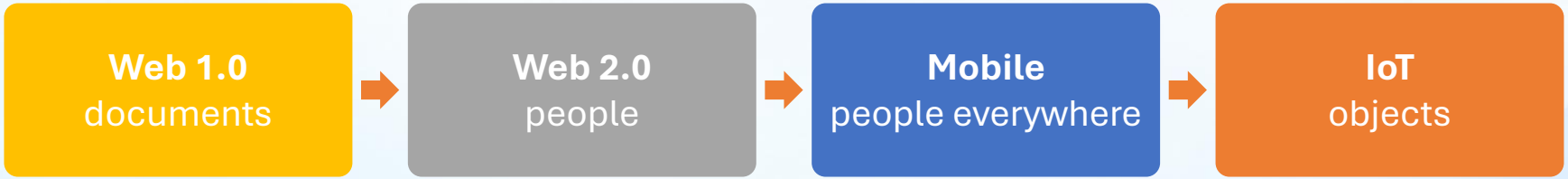
“Internet” vision
Connect objects through IP and protocols adapted to constrained devices

“Semantic” vision
Describe data so that everyone (and every machine) can understand it

Real IoT combines all three visions; objects are now the main producers of data

From the Web of documents to IoT

Web 1.0 documents → Web 2.0 people → Mobile people everywhere → IoT things



- Each stage connects something new: machines, then humans, then objects
- Today objects are the main producers of data on the Internet
- Data is no longer typed by people: it is measured automatically

IoT extends the Internet from people to the physical world

A Brief History of IoT

He believed **IoT could "turn the world into data" that could be used to make macro decisions on resource utilization.**

1982
Connected drinks machine (Carnegie Mellon)

1999
Kevin Ashton coins "Internet of Things"



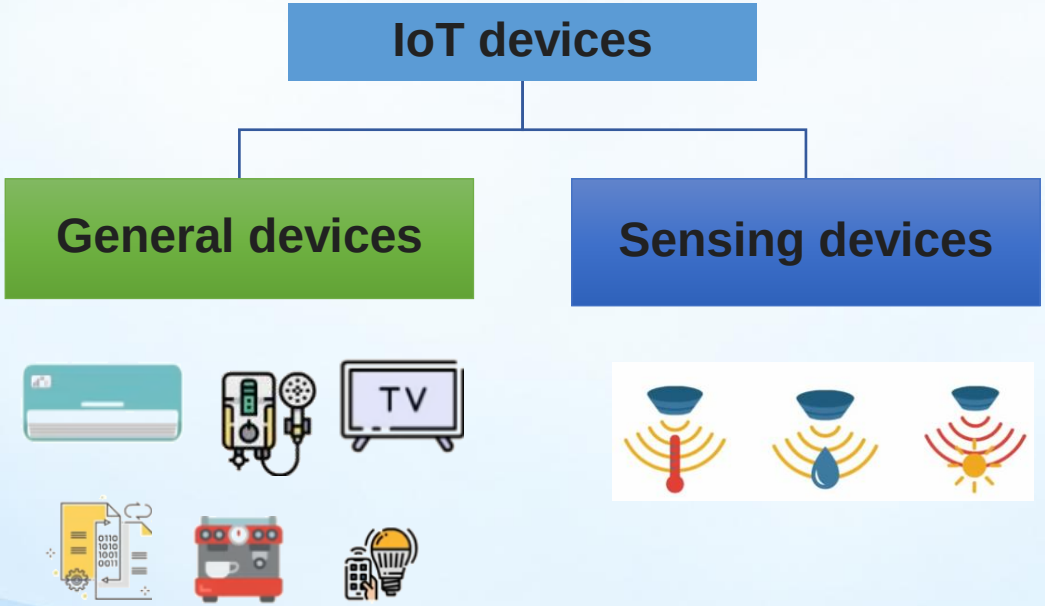
2008-09
More connected objects than humans

2010s
LPWAN, Wi-Fi/BLE, affordable cloud

2020s
5G, edge computing, embedded AI

Drivers: cheaper sensors, miniaturisation, connectivity, cloud; tens of billions of objects expected by 2030

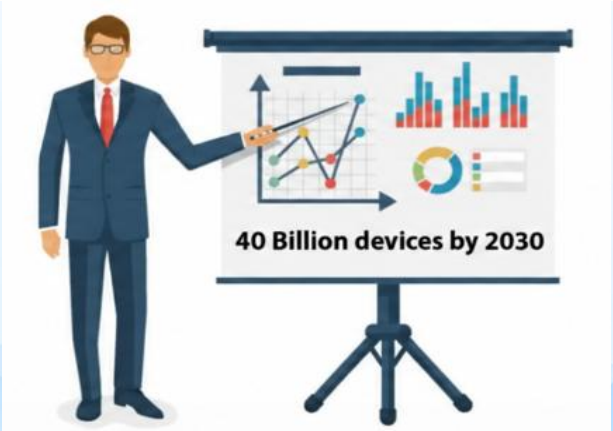
Where IoT is used ?



Growth of IoT ?



Growth of IoT



IoT Market?

The global IoT market, it is projected to be worth $1.35 trillion by 2025.



The Core Components of IoT



Sensors and Actuators

Sense real-world conditions and take action to interact with the physical world.



Connectivity

Connect devices and systems reliably and securely to exchange data.



Edge and Cloud Computing

Process and store data close to the source or in the cloud for scalability and resilience.



Analytics

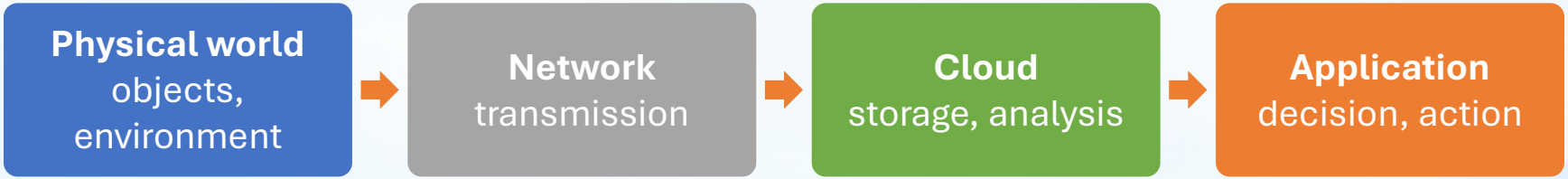
Turn data into insights with analytics, machine learning, and automation.



User Interfaces

Visualize information and control systems through intuitive interfaces.

What is the Internet of Things?



- A network of identifiable physical objects equipped with sensors, software and a connection
- Objects exchange data with each other and with services over the Internet, often without human intervention
- Goal: turn the physical world into usable information, then into action

IoT = objects + network + cloud + applications

How IoT Works



1. Sensing

Devices collect real-world data

2. Communication

Data is transmitted over networks

3. Processing

Data is processed in the cloud

4. Analysis

Data is analyzed to generate insights

5. Decision-Making

Insights inform smart decisions

6. Action

Actions are taken in the real world

Sensors and Actuators

Sensors capture real-world conditions and turn them into digital data.

Actuators receive commands and trigger physical actions in the real world.



Temperature
Measures heat and cold

Motion
Detects movement and presence

Light
Measures brightness and ambient light

Location
Tracks position and proximity

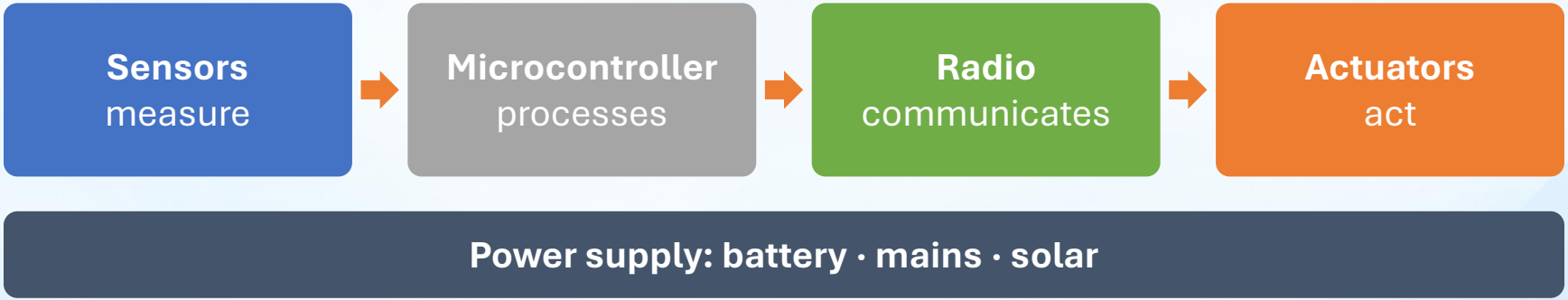
Relay / Switch
Turns devices on or off

Motor
Drives movement and mechanisms

Light
Turns lights on or off

Buzzer / Alarm
Produces sound alerts

Anatomy of a connected object



- Connected object = ordinary object + intelligence + connection
- Each object has a unique identity (address) and embedded software (firmware)

Sensor/actuator + computing + communication + energy

Sensors and actuators

	Sensor (input)	Actuator (output)
Role	Turns a physical quantity into a signal	Turns a signal into a physical action
Examples	Temperature, humidity, motion, GPS, light	Relay, motor, valve, lock, LED
Signal	Analog or digital	Electrical command
Concrete case	DHT22 on an ESP32	Irrigation solenoid valve

How IoT Works

•Various wireless radio services allow devices to connect to the Internet and to each other.
•Devices can connect using technologies like Wi-Fi, low-energy Bluetooth (BLE), Near Field Communication (NFC), ZigBee, and Z-Wave.
•Cloud services aggregate and analyze data collected from diverse connected devices.

•End users monitor system metrics and execute control actions in real time via dedicated mobile applications.
Protocol Characteristics
•Wi-Fi: Delivers high bandwidth for data-intensive devices (e.g., video streaming, gateways).
•BLE, ZigBee, & Z-Wave: Low-power, short-to-medium-range protocols designed for mesh networking and extended battery life in embedded sensor networks.
•NFC: Ultra-short-range communication used for zero-touch commissioning, secure access, and direct device pairing.

Connectivity and Communication



	Wi-Fi	Bluetooth	Cellular Networks	LPWAN	Zigbee
Range	Short	Short	Very Long	Long	Short
Power Use	Higher	Low	Higher	Very Low	Low
Typical Applications	Home devices, cameras, appliances	Wearables, audio accessories, beacons	Vehicle tracking, smart meters, POS	Asset tracking, agriculture, city infrastructure	Smart home, lighting, sensors

From Data to Insight

Turning IoT data into better decisions



1

Collect

Devices capture real-world data from sensors.

2

Transmit

Data is securely sent over networks.

3

Store

Data is stored in the cloud for reliability and scale.

4

Analyse

Data is processed to find patterns and insights.

5

Act

Insights are turned into decisions and actions.

From data to action: the value chain

Data
raw measurement

→

Information
data with context

→

Knowledge
analysis, model

→

Action
decision

Example: 32 % → dry soil → irrigation needed → open the valve

The value of IoT is in the action, not only in the measurement

Ordinary object or connected object?

Classic light bulb	Connected light bulb
• On or off with a switch • No data collected • No remote control • No programming	• Controlled by smartphone or voice • Measures its energy use • Schedules by time or presence • Software can be updated

Connected = ordinary object + intelligence + connection

Running example: the Smart Building

The Smart Building



Sensors measure

- Temperature and humidity
- Light level
- Air quality
- Occupancy
- Energy consumption

The cloud analyses

- Processes real-time data
- Detects patterns and anomalies
- Makes decisions (e.g. adjust heating, lighting, ventilation)
- Predicts maintenance needs

Actuators correct

- HVAC system (heating / cooling / ventilation)
- Lighting
- Shades / blinds
- Water / irrigation
- Elevators

Benefits

Comfort

- Ideal temperature
- Better air quality
- Appropriate lighting

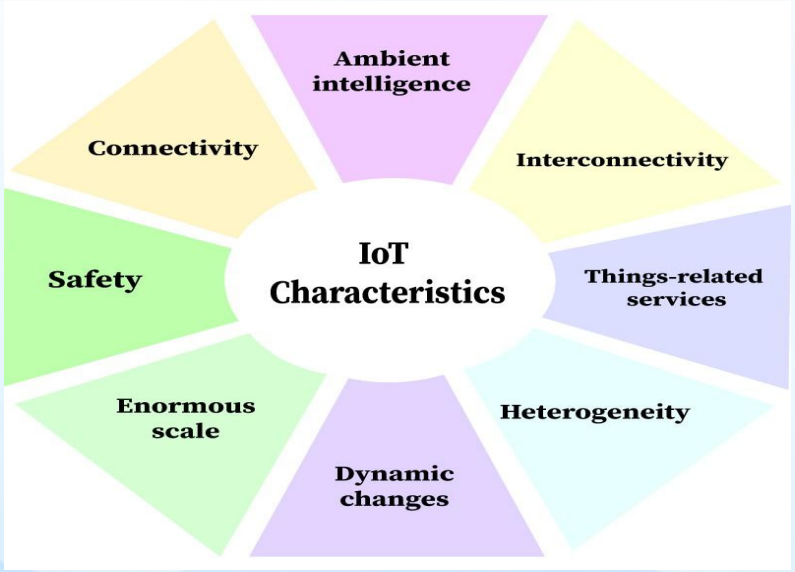
Energy savings

- Optimized heating/cooling
- Smart lighting
- Reduced waste

Predictive maintenance

- Early fault detection
- Less downtime
- Lower costs

2. Characteristics





Eight key characteristics of IoT



1 Connectivity

Objects communicate over a network

2 Identification

Unique identity for each object

3 Sensing

Perception of the environment

4 Intelligence

Local or remote processing

5 Heterogeneity

Varied hardware and protocols

6 Scalability

Thousands to millions of objects

7 Autonomy

Works without humans

8 Security

Protect objects and data

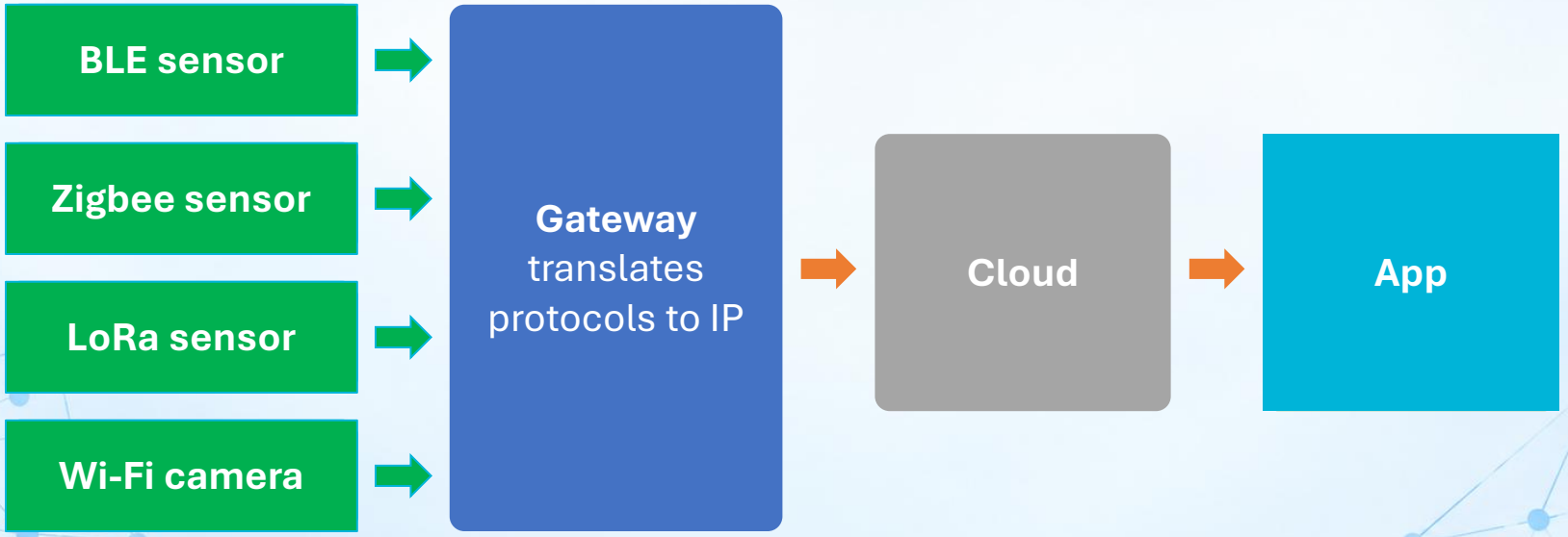
These traits distinguish an IoT system from an ordinary network

Identification: who is talking?

Identifier	Layer	Example
MAC address	Link	A4:CF:12:34:56:78
IP address	Network	IPv4 or IPv6
EPC / RFID	Tag	Tracking a parcel
URI / topic	Application	building/room1/temp

Without a unique identifier, we cannot know which object is speaking

Heterogeneity: the gateway unifies



The gateway translates heterogeneous protocols into IP

Resource constraints of IoT devices

Resource	Server / PC	Typical IoT object
Processor	Several GHz	A few MHz to 240 MHz
RAM	GB	A few KB to hundreds of KB
Energy	Mains power	Battery, months or years
Network	Fast, stable	Low rate, frequent losses
Cost	High	A few euros

Consequence: lightweight protocols, frugal code, deep sleep

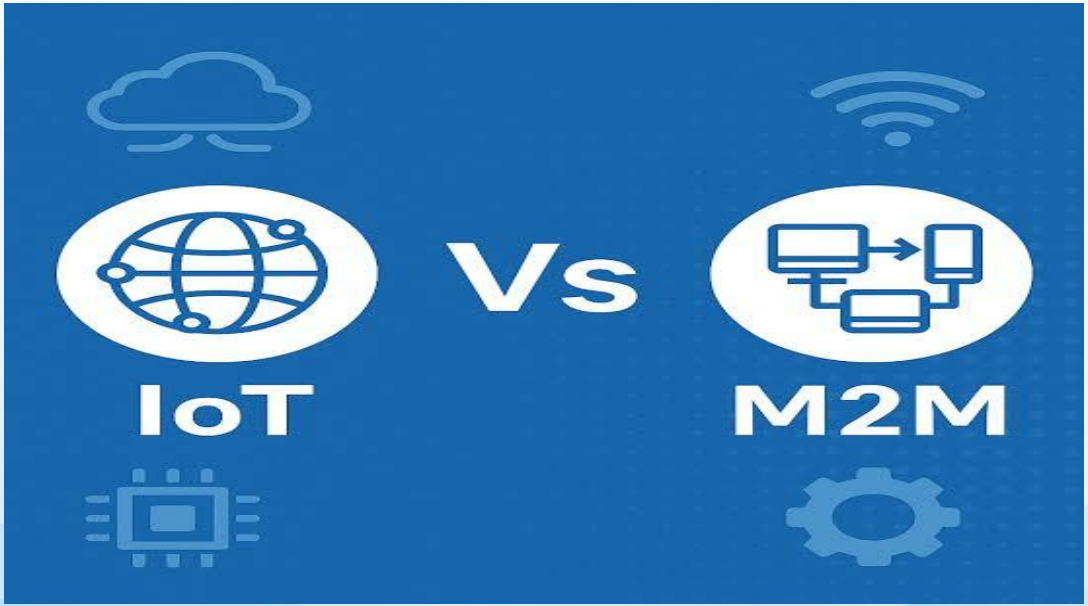
Scalability, dynamism and autonomy

Scalability
The system must handle millions of objects without degrading the service

Dynamism
Mobile objects: they join, leave and change networks

Autonomy
Self-configuration, self-healing, local decisions

3. IoT vs M2M



M2M: Machine-to-Machine

Machine
meter, vending machine

→

Network
cellular or wired

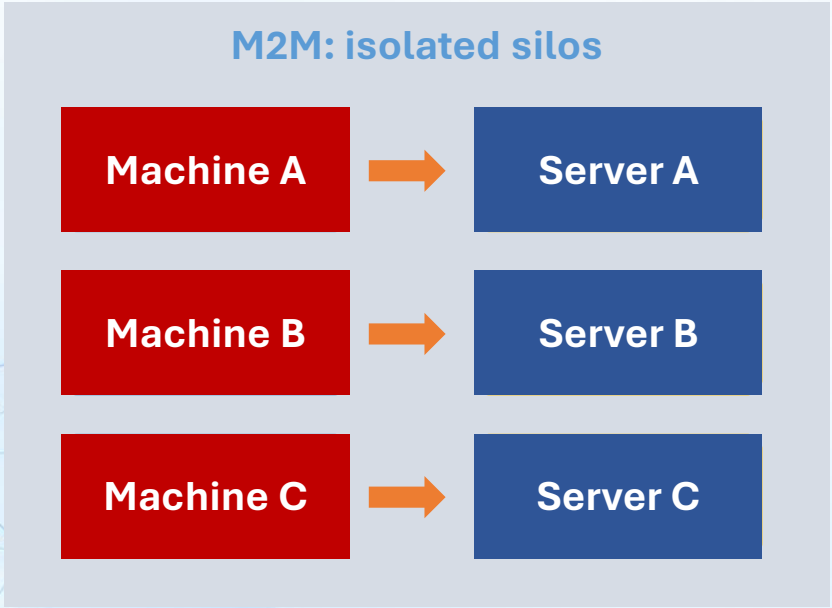
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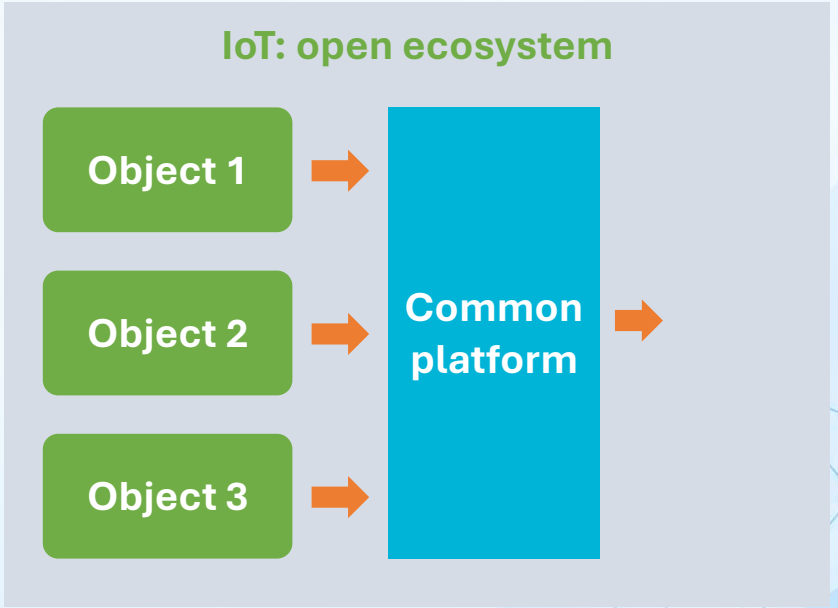
Dedicated server
single application

- Direct communication between machines, often point-to-point
- Vertical solutions: proprietary and closed
- Little or no human interaction

M2M is the ancestor of IoT, and now a subset of it

M2M vs IoT: silos or ecosystem?

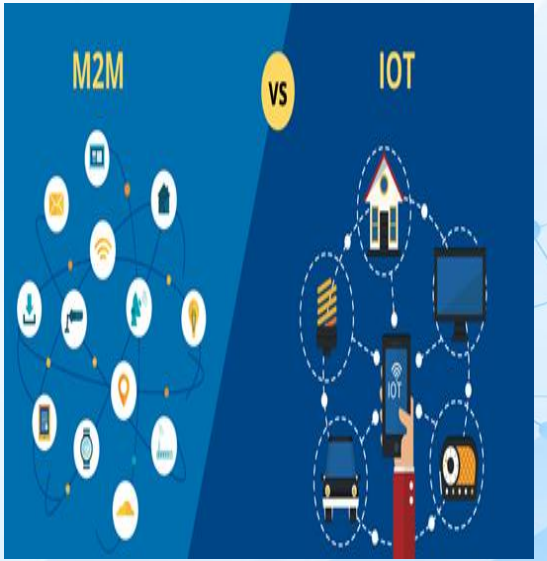




M2M is a subset of IoT

Detailed comparison: IoT vs M2M

Criterion	M2M	IoT
Connection	Point-to-point, cellular or wired	Internet, IP, multi-network
Architecture	Vertical silos	Open horizontal platform
Data	Simple exchange	Big data, analytics, AI
Standards	Often proprietary	Open (MQTT, CoAP)
Scale	Limited	Massive
Human interaction	Low	High (applications)



Example: the energy meter

M2M meter

Meter
sends index

→

Utility server
billing

IoT meter

Meter
fine-grained data

→

Platform
analytics

→

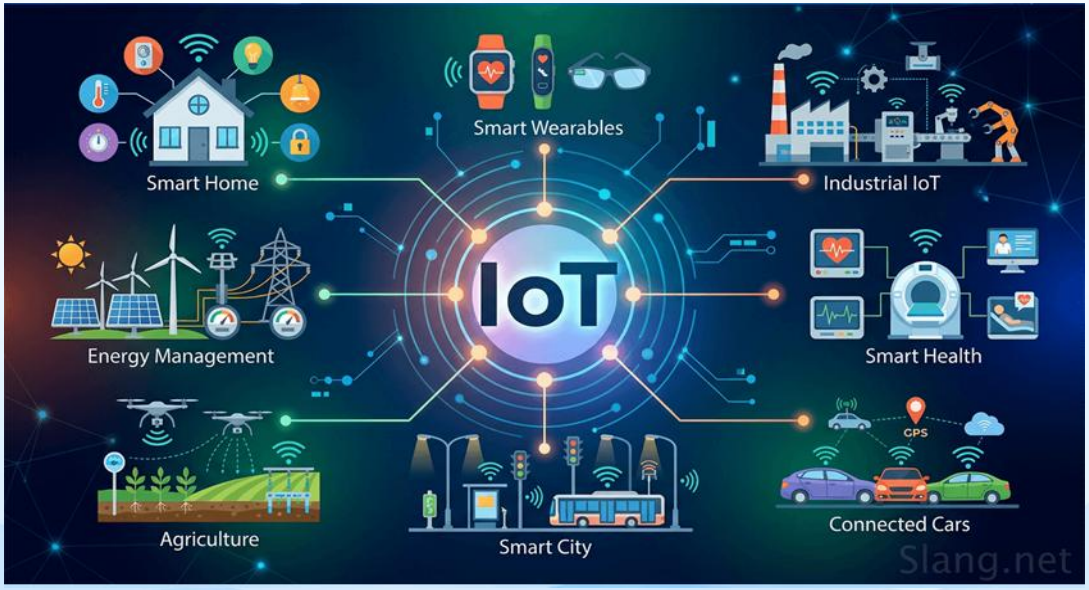
Customer app
advice

M2M: one meter, one use. IoT: same data, several uses and services.

Example: the energy meter



4. Application Domains



Where IoT is used ?

Common domains differ in scale, latency, reliability, and risk



- **Consumer**
  homes, wearables, appliances
- **Industrial IoT**
  machines, plants, logistics
- **Health**
  monitoring, devices, care pathways
- **Smart cities**
  traffic, energy, waste, safety
- **Agriculture**
  soil, water, livestock, equipment

Domain constraints define architecture: a smartwatch optimizes battery life, while a factory prioritizes deterministic uptime and safety.

Panorama of application domains

Smart home
Thermostats, security, lighting

Smart city
Street lights, parking, waste

Health
Remote patient monitoring

Agriculture
Irrigation, crops, livestock

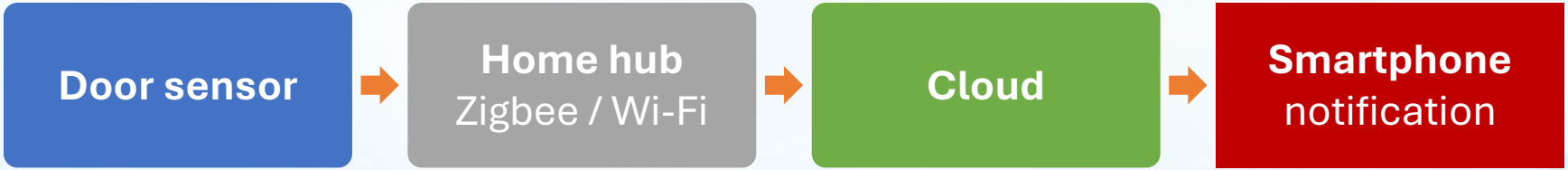
Industry 4.0
Predictive maintenance

Transport
Fleets, connected vehicles

Energy
Smart grids

Retail
Inventory, logistics

Smart home



- Typical scenario: “if nobody is home, switch off lights and heating”
- Devices: lighting, smart plug, thermostat, smoke detector, lock, camera
- Stakes: privacy and compatibility between brands
- Common protocols: Wi-Fi, Zigbee, BLE, MQTT

IoT in the Smart Home

Connected devices that make daily life smarter, simpler, and more efficient.



Smart Lighting

Adjust brightness and color, set schedules, and save energy.



Thermostats

Learn your preferences and optimize comfort and efficiency.



Security Cameras

Monitor in real time, get alerts, and review from anywhere.



Smart Locks

Lock, unlock, and manage access remotely with peace of mind.



Voice Assistants

Hands-free control of devices, routines, and information.



Convenience

Control your home anytime, anywhere.

Energy Efficiency

Reduce waste and lower utility costs.

Automation

Routines and scenes that work for you.

Smart city

Street lighting
Adaptive brightness saves energy

Air quality
Pollution sensors in real time

Parking
Free spots shown to drivers

Traffic
Adaptive traffic lights

Waste
Bins signal when full

Urban platform
One platform for all services

Many sensor types feed one shared urban platform


SMART
CITY

Connected health

Patient
wearable sensors

→

Smartphone
BLE relay

→

Cloud
analysis

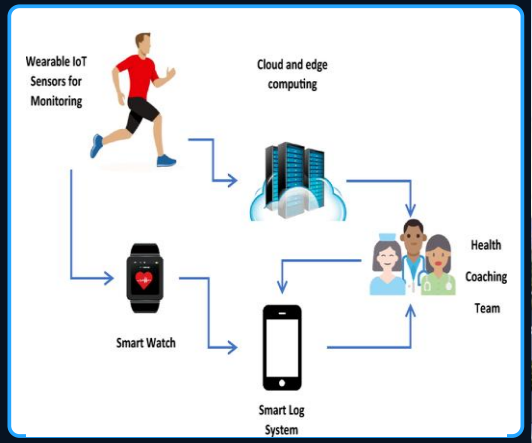
→

Doctor
alert, follow-up

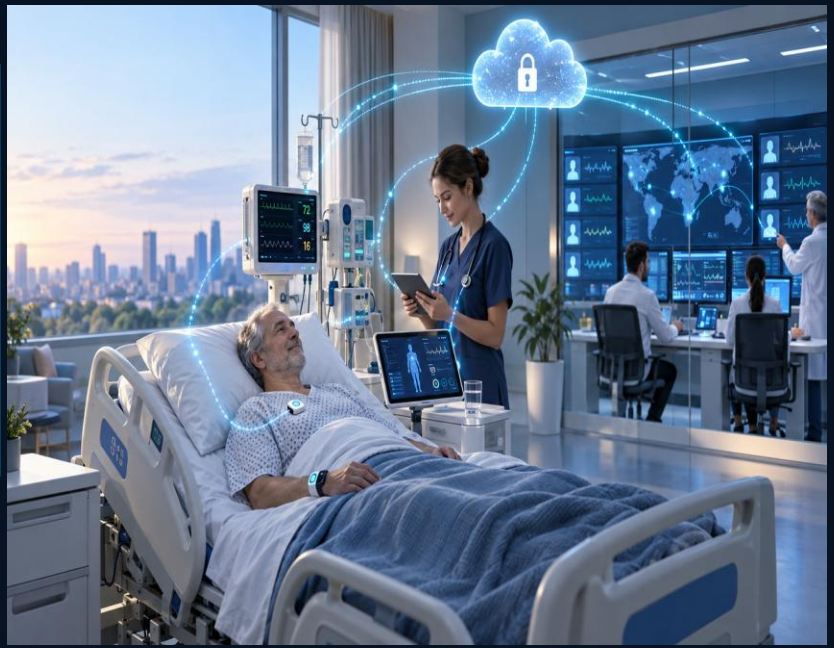
- Measurements: heart rate, glucose, oxygen saturation
- Automatic alerts when thresholds are crossed
- Requirements: reliability, confidentiality, low latency for alerts

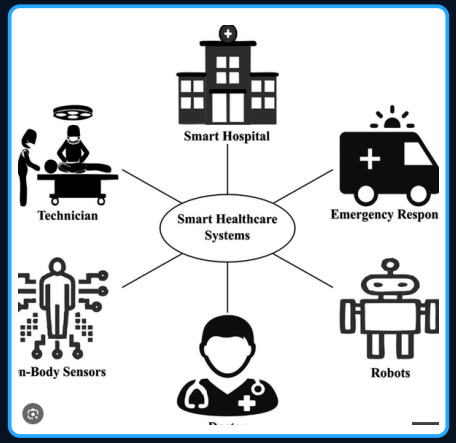
Scenario: Health Care

Wearables, medical devices, edge/cloud processing, and clinical teams form a continuous monitoring loop.



Remote patient monitoring pipeline

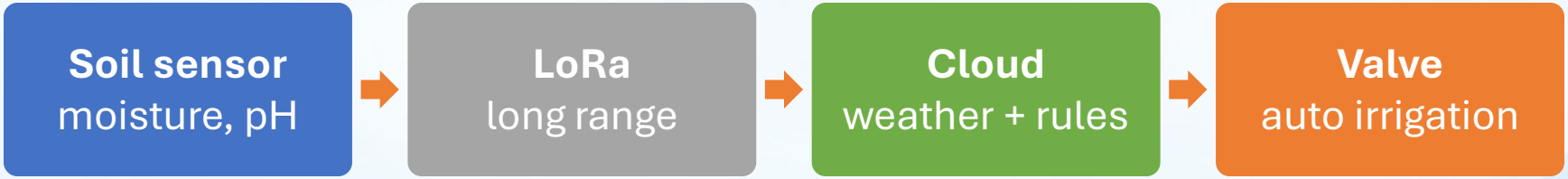




Smart healthcare service ecosystem

Key insight: IoT improves healthcare by enabling continuous data capture, real-time analytics, earlier intervention, and coordinated emergency response.

Smart agriculture



- Needs: range of several km, battery life of several years
- Suitable technology: LoRaWAN
- Benefits: water savings, better yield

Smart Farming

IoT Application

CONNECTED FIELDS • HEALTHY FOOD • A BRIGHTER TOMORROW

SMART
TECHNOLOGY
GREENER
TOMORROW

Crop Monitoring

Sensors track soil moisture, temperature, and crop health.

Smart Irrigation

Automated watering reduces waste and improves efficiency.

Livestock Tracking

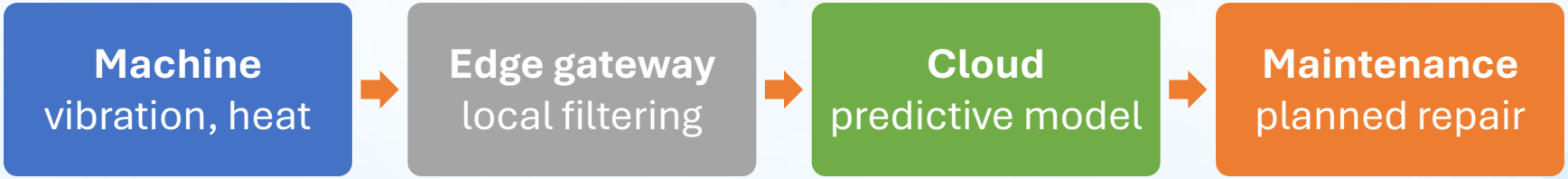
Wearables and GPS help monitor animal health and movement.

Data Analytics

Cloud platforms support weather forecasting and yield prediction.



Industry 4.0: predictive maintenance



- Repair before the breakdown, not after
- Requirements: robustness, real time, security

Less downtime, lower maintenance cost

Industrial Internet of Things

(IIoT)

Connected industrial assets use sensors, edge gateways, cloud analytics, and automated control to turn factory data into operational decisions.

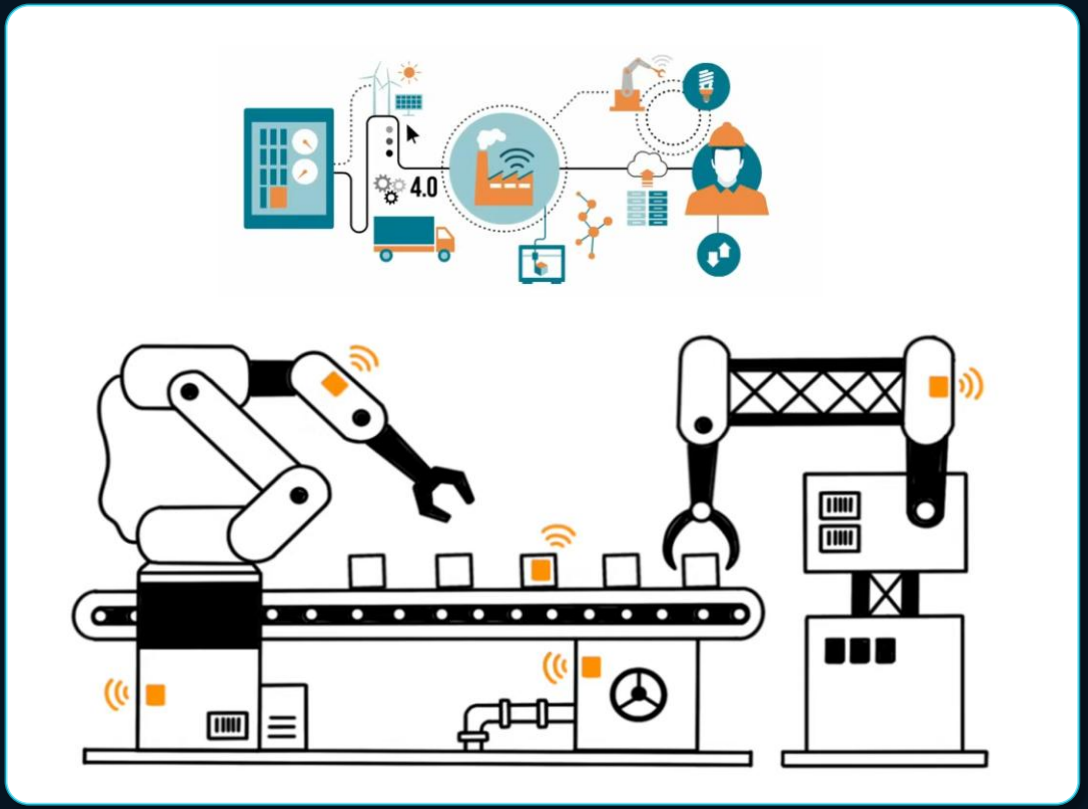
IIoT functional layers

01 Machine sensing
Motors, robots, conveyor lines, and industrial equipment generate real-time signals.

02 Edge control
Local gateways filter, secure, and respond to latency-sensitive events.

03 Cloud intelligence
Analytics and dashboards identify anomalies, efficiency gaps, and maintenance needs.

04 Action loop
Insights trigger process optimization, alerts, and safer automated operations.



Key outcomes: predictive maintenance • reduced downtime • higher throughput • improved safety

The IoT Ecosystem

How connected things create real-world value



Devices

Sensors and actuators collect real-world data

Connectivity

Data is securely transmitted over networks to the cloud

Data Processing

Data is stored, analyzed, and turned into insights

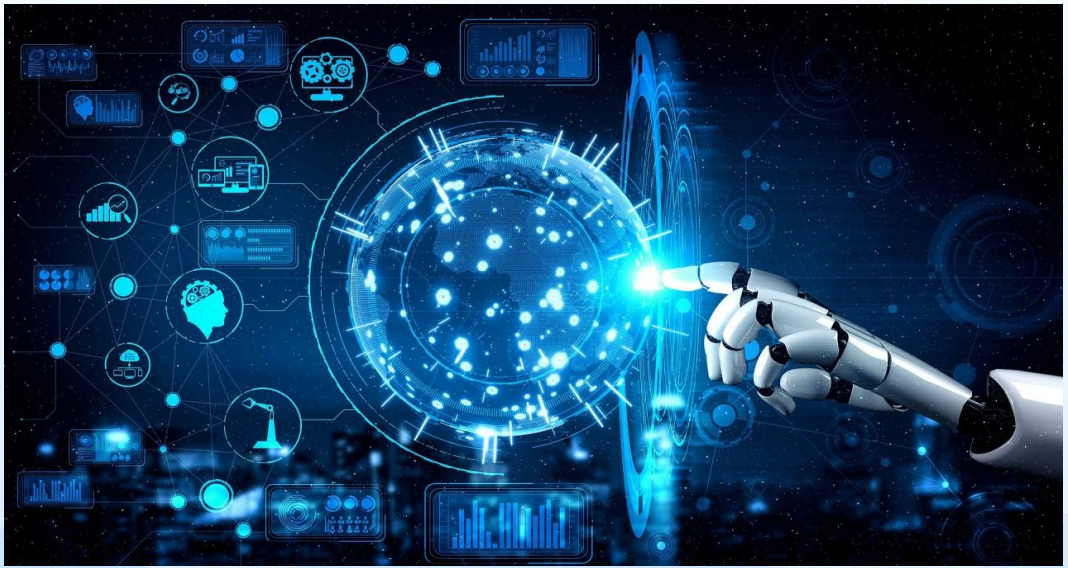
Applications

Insights power apps and automation to drive actions

Users

People and systems use applications to make better decisions

5. Enabling Technologies



Five key enabling technologies

Wireless sensor networks
Distributed data collection

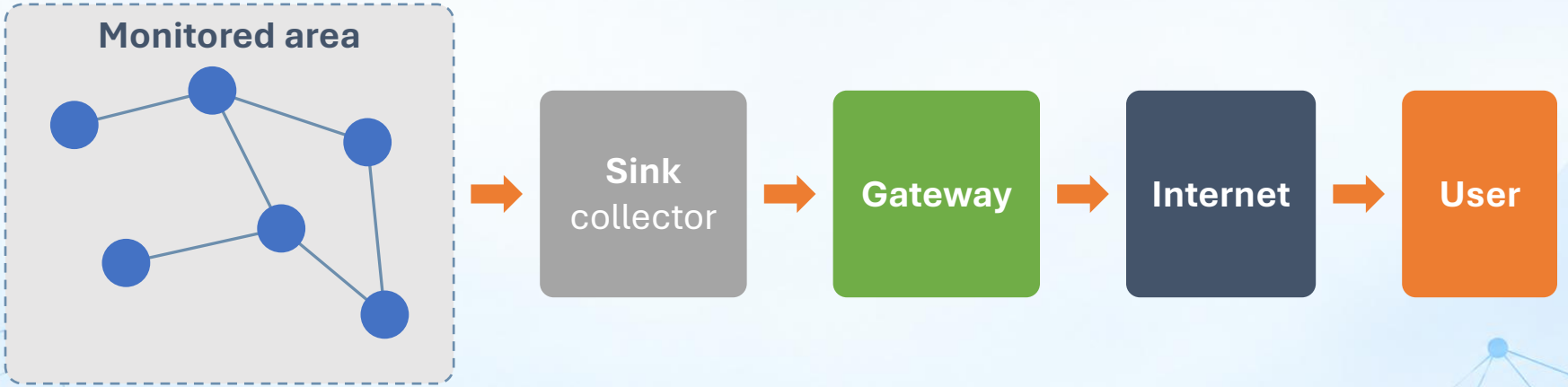
Embedded systems
The “brain” of the object

Communication protocols
Light and reliable exchange

Cloud computing
Storage and compute on demand

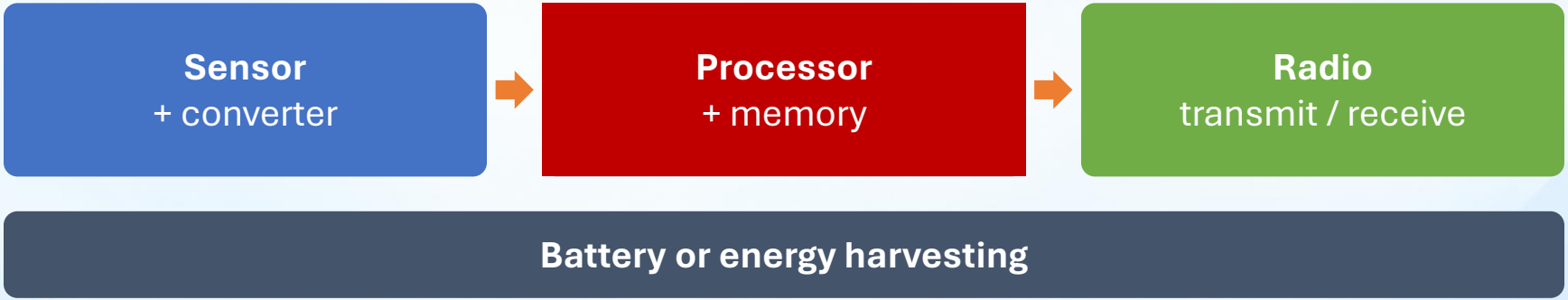
Big data analytics
Extracting value from data

WSN: architecture of a sensor network



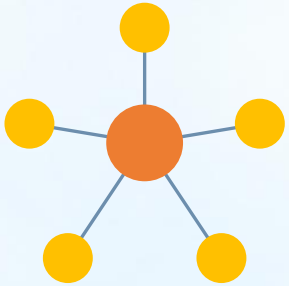
- Each node: sensor + processor/memory + radio + battery
- Roles: measure, aggregate, relay neighbours' data (multi-hop routing to the sink)
- Constraints: energy, short radio range, small memory

WSN: anatomy of a sensor node



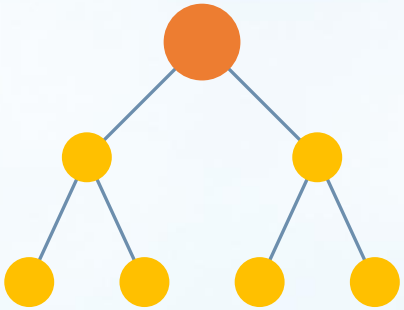
- Roles: measure, aggregate, relay neighbours' data
- Constraints: limited energy, short radio range, small memory

WSN topologies



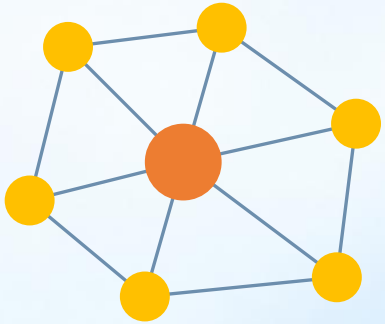
Star

Simple, but the centre is a single point of failure



Tree

Hierarchical: good range/cost compromise

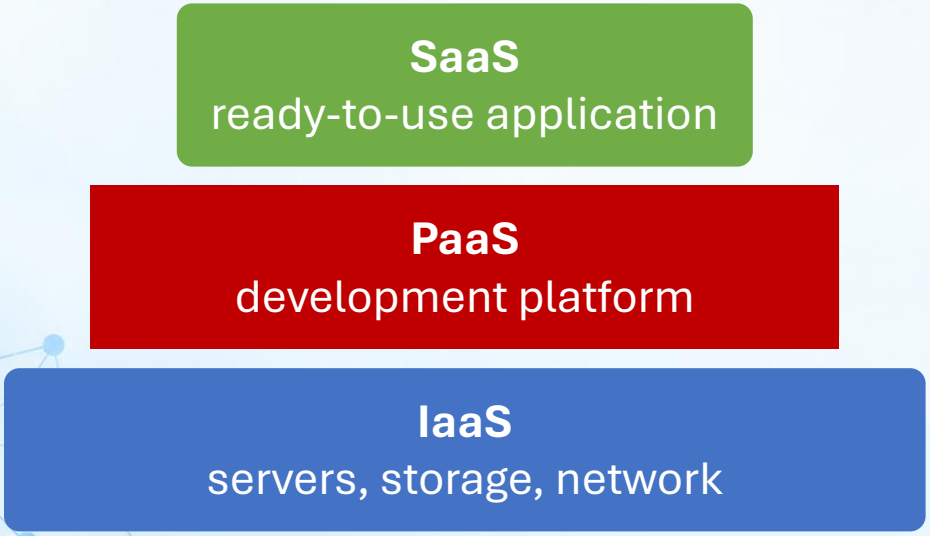


Mesh

Multiple paths: robust, routing more complex

Choice depends on range, energy and required reliability

Cloud computing: service models



- IaaS: you manage OS and apps
- PaaS: you manage only code and data
- SaaS: you simply use the service
- Deployment: public, private or hybrid
- Role in IoT: massive storage, elastic compute, dashboards
- Examples: AWS IoT, Azure IoT Hub, Google Cloud

Edge, fog and cloud: where to compute?

Devices
sensors, actuators

→

Edge / fog
local processing

→

Cloud
global analysis

Edge / fog
Low latency, works offline, less data sent

Cloud
Heavy analytics, long-term storage, global view

The cloud is not mandatory: process locally when speed or privacy matters

Big data analytics: the 5 Vs and the pipeline

Volume
massive quantity

Velocity
fast flow

Variety
many formats

Veracity
reliable readings

Value
useful decisions

Collect → **Store** → **Process** → **Analyse** → **Visualise**

IoT data = high volume, high velocity, high variety — value comes from analysis

Four levels of analytics

Descriptive: what happened?
e.g. this week's consumption

Diagnostic: why?
e.g. peak caused by air conditioning

Predictive: what will happen?
e.g. likely failure

Prescriptive: what to do?
e.g. lower A/C at 2 pm

Communication protocols: the stack





Application
HTTP, CoAP, MQTT

Transport
TCP, UDP

Network
IPv6, 6LoWPAN

Link / physical
Wi-Fi, BLE, 802.15.4, LoRaWAN

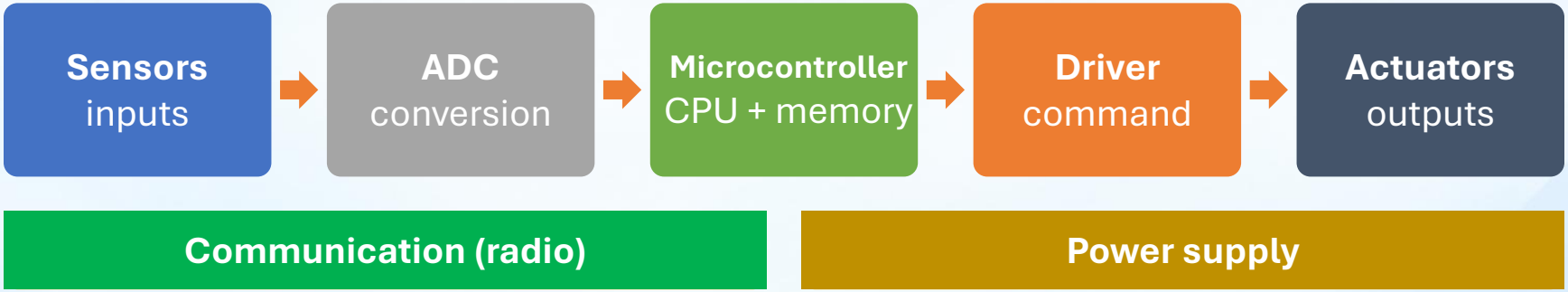
- Each layer hides the details of the layer below
- Choice depends on range, data rate and energy
- Details (MQTT, CoAP…) come in Chapter 2

Choosing a communication technology

Need	Suitable technology
Few metres, very low energy	Bluetooth LE
High data rate, mains power	Wi-Fi or Ethernet
Meshed smart home	Zigbee / Thread (802.15.4)
Several km, multi-year battery	LoRaWAN
Mobility, wide coverage	4G / 5G (LTE-M, NB-IoT)

No technology is perfect: trade off range, data rate and energy

Embedded systems: definition



- A computer system dedicated to a specific task, built into a product
- Constraints: memory, energy, cost, size, reliability, timing

Embedded software architecture





Application
business logic

RTOS / libraries
scheduling, drivers

Hardware
MCU, memory, peripherals

- Firmware = application + system software
- An RTOS (e.g. FreeRTOS) manages tasks with deadlines
- Simple objects sometimes run without any OS

Interaction models: timing constraints

No time constraint
A delay is acceptable.
E.g. hourly temperature reading.

Soft real time
A delay degrades quality.
E.g. audio streaming.

Hard real time
A delay is a failure. E.g. airbag, braking.

Key question: response time ≤ deadline?

Embedded design approaches

Top-down ↓

- Needs
- Architecture
- Components

Start from needs and go down to components.

Bottom-up ↑

- Needs
- Architecture
- Components

Start from existing components and assemble.

Mixed ↕

- Needs
- Architecture
- Components

Both directions meet: reuse + needs.

Hardware/software co-design (codesign)



- If constraints are not met: go back to partitioning (iterate)
- Goal: optimise performance, cost and energy together
- Hardware is fast but fixed; software is flexible but slower

Codesign example: connected thermostat

Function	Choice	Reason
Sensor reading	Hardware (ADC)	Fast, low energy
Measurement filtering	Software	Easy to modify
Encryption (AES)	Hardware (accelerator)	Fast, efficient
Interface and rules	Software	Upgradable by update

6. Challenges



Key Issues & Challenges

What happens to the collected data?

How can we prevent companies from selling our data?



Where is it stored?

Key Issues & Challenges in IoT

Security risks
from data breaches and unauthorized access

Privacy concerns
around data collection and user consent

Interoperability challenges
across multiple standards and platforms

Scalability demands
in managing billions of connected devices





The big challenges of IoT



Security
Attacks, missing updates

Privacy
Sensitive personal data

Interoperability
Multiple standards

Energy
Limited battery

Scalability
Millions of objects

Reliability
Failures, unstable networks

Cost
Cheap objects

Standardisation
Fragmented norms

Security

Many IoT devices ship with minimal built-in protection, making them attractive targets.



Why it matters

- Devices often use default or hardcoded passwords that are rarely changed
- Manufacturers may stop issuing firmware updates soon after release
- Compromised devices can be recruited into large-scale botnets
- Physical accessibility of devices increases tampering risk

REAL-WORLD EXAMPLE

The 2016 Mirai botnet hijacked hundreds of thousands of insecure cameras and routers to launch one of the largest DDoS attacks recorded, disrupting major internet services.

Security: threats by layer

Layer	Threats
Perception	Device theft, tag cloning, radio jamming
Network	Eavesdropping, man-in-the-middle, denial of service
Application / cloud	Web flaws, unauthorised access, data theft

Real case: the Mirai botnet (2016)

Scan
looks for devices

→

Default passwords
access

→

Botnet
thousands of devices

→

DDoS attack
services disrupted

- Targets: poorly protected cameras and routers
- Lesson: change default credentials and update firmware (OTA)

Security: good practices

Encryption
Protect data in transit and at rest

Authentication
Verify every device and user

Updates (OTA)
Patch firmware remotely and regularly

No default passwords
Change credentials at first use

Secure every layer: object, network, cloud

Interoperability

A fragmented landscape of standards keeps devices from working together smoothly.



Why it matters

- Competing protocols include Wi-Fi, Bluetooth, Zigbee, and Z-Wave
- Manufacturers frequently build closed ecosystems that resist integration
- Consumers may need multiple hubs or apps to manage one home
- Lack of shared standards raises costs and slows innovation

AN EMERGING FIX

The Matter standard, backed by Amazon, Apple, Google, and others, aims to let smart home devices work together regardless of brand - though adoption is still uneven.

Interoperability and standardization



- Without a standard, each brand needs its own app
- Solutions: open protocols, APIs, common semantics

Privacy

Always-on sensors collect intimate details about daily life, often continuously.



Why it matters

- Smart devices can capture location, voice, video, and behavioral patterns
- Consent is often buried in lengthy terms of service few people read
- Data may be shared with or sold to third parties for advertising
- Aggregated data can reveal sensitive patterns even when anonymized

CONSIDER

A single smart home can generate a detailed timeline of when residents wake, leave, and sleep — information with real value to advertisers, insurers, and bad actors alike.

Privacy, reliability and cost

Privacy
Minimise collected data, encrypt, inform the user

Reliability
Redundancy, automatic reconnection, monitoring

Cost
A few euros per object: trade-off with security and lifetime

Scalability

Billions of devices generate more data than traditional infrastructure was built for.



Why it matters

- Continuous sensor data creates enormous storage and bandwidth demands
- Centralized cloud processing can introduce latency for time-sensitive tasks
- Network addressing had to evolve (IPv4 to IPv6) to support more devices
- Edge computing is increasingly used to process data closer to its source

IN PRACTICE

A single autonomous vehicle can generate several terabytes of sensor data per day - far too much to send to the cloud in real time, pushing processing to the edge.

Thank you for your attention

